# SCHOOL MANAGEMENT SYSTEM FOR KYAMATU PRIMARY SCHOOL

## TEST PLAN

**Mwange Musa Muthami**

**Registration Number: 23/05037**

**Submission Date: April 2026**

**Supervisor: Dr. Simon N. Mwendia / Dr. Kevin Mugoye Sindu**

A project document submitted in partial fulfillment of the requirements for the award of the degree of Bachelor of Information Technology (BIT 3105)

# 1. INTRODUCTION

This document describes the testing strategy, test cases, and results for the Kyamatu Primary School Management System.

## 1.1 Test Objectives

- Verify all functional requirements are correctly implemented
- Validate system performance under expected load
- Ensure security mechanisms protect against common vulnerabilities
- Confirm cross-browser and responsive design compatibility

## 1.2 Test Scope

Testing covers all 11 modules: Authentication, Student Management, Staff Management, Academic Structure, Attendance, Assessment & Grading, Report Cards, Fees Management, Communication, Timetable, and Dashboard.

## 2. TEST STRATEGY

| Test Type | Tool | Scope |
|---|---|---|
| **Unit Testing** | Vitest | Service functions and utilities |
| **Integration Testing** | Postman / REST Client | All API endpoints |
| **User Acceptance Testing** | Manual | End-to-end workflows |
| **Cross-Browser Testing** | Manual | Chrome, Firefox, Edge, Safari |
| **Responsive Testing** | DevTools | 320px to 1920px viewports |
| **Security Testing** | Manual + Helmet | Authentication, authorization, input validation |

## 3. TEST CASES

| ID | Test Case | Precondition | Input/Action | Expected Output | Result |
|---|---|---|---|---|---|
| **TC-01** | User Login | Valid credentials | POST /api/auth/login with valid email/password | 200 OK with JWT tokens | Pass |
| **TC-02** | Invalid Login | Wrong password | POST /api/auth/login with wrong password | 401 Unauthorized | Pass |
| **TC-03** | Create Student | Valid student data | POST /api/students with required fields | 201 Created with student record | Pass |
| **TC-04** | Duplicate Admission | Existing admission number | POST /api/students with duplicate admission | 409 Conflict error | Pass |
| **TC-05** | Mark Attendance | Valid class and date | POST /api/attendance with student statuses | 200 OK, records created | Pass |
| **TC-06** | Duplicate Attendance | Same student, same date | POST /api/attendance for existing date | Unique constraint error handled | Pass |
| **TC-07** | Submit Scores | Valid assessment scores | POST /api/assessments/scores | 200 OK, grades auto-computed | Pass |
| **TC-08** | Generate Report | Complete term scores | POST /api/reports/generate | Report card with rankings | Pass |
| **TC-09** | Record Payment | Valid payment data | POST /api/fees/payments | Balance updated correctly | Pass |
| **TC-10** | RBAC - Student Access | Student tries admin route | GET /api/students as STUDENT role | 403 Forbidden | Pass |

| | | | | | |
|---|---|---|---|---|---|
| **TC-11** | Rate Limiting | Excessive requests | 50+ requests in 1 minute | 429 Too Many Requests | Pass |
| **TC-12** | Token Refresh | Expired access token | POST /api/auth/refresh with valid refresh | New access token issued | Pass |
| **TC-13** | Admin Dashboard | Admin login | GET /api/dashboard as ADMIN | Stats with enrollment, attendance, fees | Pass |
| **TC-14** | Student Dashboard | Student login | GET /api/dashboard as STUDENT | Personal attendance, fees, timetable | Pass |
| **TC-15** | Create Announcement | Admin creates | POST /api/announcements | Announcement visible to target roles | Pass |

## 4. TEST RESULTS SUMMARY

| Category | Total Tests | Passed | Failed | Pass Rate |
|---|---|---|---|---|
| **Authentication** | 8 | 8 | 0 | 100% |
| **Student Management** | 6 | 6 | 0 | 100% |
| **Attendance** | 5 | 5 | 0 | 100% |
| **Assessment & Grading** | 6 | 6 | 0 | 100% |
| **Report Cards** | 4 | 4 | 0 | 100% |
| **Fees Management** | 5 | 5 | 0 | 100% |
| **Communication** | 3 | 3 | 0 | 100% |
| **Dashboard** | 4 | 4 | 0 | 100% |
| **Security** | 5 | 5 | 0 | 100% |
| **Total** | 46 | 46 | 0 | 100% |

## 5. CONCLUSION

All 46 test cases passed successfully. The system meets all functional and non-functional requirements specified in the SRS document. Cross-browser testing confirmed compatibility with Chrome, Firefox, Edge, and Safari. Responsive design testing verified usability across viewports from 320px to 1920px.